Dr. Martin JACQUET

Postdoctoral Researcher

PhD in Robotics
Engineer in Computer Sciences and Applied Mathematics

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Experiences & Education

Sep 2025 – present

Postdoctoral Researcher, at INRIA - RAINBOW team, Rennes, France
Perception-based physical interaction with aerial robots
With Dr. Marco Tognon – ANR-FlyHandyBot project
- Nonlinear control for vision-constrained tasks
- Hybrid/Stochastic NMPC, MPPI
- Integration of onboard lidar-inertial state estimator
- ROS2, GenoM; Python, C++ programming

2020 – present

Temporary Lecturer
Qualification for the position of Maître de Conférences (section CNU 61) obtained in 2026
- At ENS Rennes (M1, M2):
  - (2027) (scheduled) 32 hours of lectures and practicals – Nonlinear control
  - (2026) 28 hours of lectures and practicals – Nonlinear control
  - (2025) 6 hours of lectures – Aerial robotics: modeling and control
- At NTNU (M1):
  - (2023) 10 hours of lectures and practicals – ROS development
- At INP-ENSEEIHT (L3, M1):
  - (2021) 31 hours of seminars and practicals – C programming, Android programming
  - (2020) 21 hours of practicals – Android programming

Feb 2023 – Jun 2025

Postdoctoral Researcher, at NTNU - Autonomous Robots Lab, Trondheim, Norway
Perception-based control methods for collision-free navigation in unknown, complex and cluttered environments
With Prof. Kostas Alexis – EU-Digiforest project
- Deep learning for latent state/environment representations
- Interplaying neural methods with principled control methods (NMPC, CBF) for provable safety
- Software integration and Autonomy Stack development
- Neural world models, Reinforcement Learning
- ROS2, ROS, PX4; python, C++ programming
- Robotics field-testing campaigns: hardware deployment, mission operator in forest environments

Jan 2019 – Oct 2022

PhD in Robotics, at LAAS-CNRS - RIS group, Toulouse, France
Methods for Online Predictive Control of Multi-rotor Aerial Robots with Perception-driven Tasks subject to Sensing and Actuation Constraints
Advised by Prof. Antonio Franchi – ARN-MuRoPhen project
- Perception-constrained NMPC, Multi Aerial Robot Systems
- Active Perception methods, Observation uncertainty estimation
- ROS, GenoM; Python, Matlab and C++ programming
- Involvement in the 2020 MBZIRC robotics competition: perception-related software programming
  →**Bronze medal** in the Grand Challenge competition

Mar 2017 – Jun 2018

Research Engineer at LAAS-CNRS - RAP group, Toulouse, France
- Collaborative project with the French Wine Institute (Gaillac) and Naïo Technology (Toulouse)
- Deep Learning-based detection of trunks and leaf diseases on wine images and videos
- Semantic segmentation, object detection, spatio-temporal classification
- Python and C++ programming, libraries Keras and TensorFlow

Summer 2016

6 months engineering internship at ATOS, Toulouse, France
Detection and tracking of icebergs in satellite images
- Semantic segmentation, classifications, temporal tracking
- C++ programming, web technologies (API development, SQL database, Docker, Kubernetes)

References and additional information available upon request

Summer 2015 — **3 months research internship at IRIT**, Toulouse, France
Search of symmetries for 3D-meshes compression
- Transposition of 2D algorithms in 3D for feature extraction (SIFT)
- Statistical evaluation methodology
- Matlab programming

Summer 2014 — **Work training period in a Japanese hotel, as a cleaning person**
Kamogawa Hotel Mikazuki, Kominato, Chiba, Japan

2013 – 2016 — **Engineering degree, at INP-ENSEEIHT**, Toulouse, France
Computer sciences and applied mathematics department
- Audiovisual data processing - Computer vision - 3D Modeling - Virtual Reality
- Mathematics - Statistics/Probabilities - Linear algebra - Optimization
- Computer sciences - Software programming
- School semester at Polytechnique Montreal, Canada; Courses in multimedia technologies
- 2 months master's project : Building a 3D model of a scene from a flow of RGBD data

2011 – 2013 — **Preparatory school, PSI* (Physics and Engineer Sciences)**
Lycée Champollion, Grenoble, France

2011 — **Scientific Baccalauréat**
Lycée Pierre Termier, Grenoble, France

Scientific Publications

Citation count and data as of 20/10/2026
Total citations: 235
h-index: 8

Journal Publications

2nd revision ongoing at T-RO, 2026 — M. Harms, **M. Jacquet**, K. Alexis, *Safe Reactive 3D Navigation of Multirotor UAVs with Neural Control Barrier Functions*

under review at IJRR, 2026 –Cited: 1 — M. Dharmadhikari, N. Khedekar, M. Kulkarni, M. Nissov, **M. Jacquet**, *et al.*, *The Unified Autonomy Stack: Toward a Blueprint for Generalizable Robot Autonomy*

IJRR, 2025 –Cited: 12 — **M. Jacquet**, M. Harms, K. Alexis, *Neural NMPC through Signed Distance Field Encoding for Collision Avoidance*

RAL, 2022 –Cited: 22 — **M. Jacquet**, M. Kivits, H. Das, A. Franchi, *Motor-level N-MPC for Cooperative Active Perception with Multiple Heterogeneous UAVs*

RAL, 2021 –Cited: 44 — **M. Jacquet**, A. Franchi, *Motor and Perception Constrained NMPC for Torque-Controlled Generic Aerial Vehicles*
– **Finalist for Best Paper Award** in Unmanned Aerial Vehicles at ICRA'21

Book Chapters

Springer, 2026 — M. Camurri, E. Tomelleri, M. Mattamala, S. Barbas Laina, **M. Jacquet**, *et al.*, *DigiForest: Digital Analytics and Robotics for Sustainable Forestry*, in *Recent Advances in Robotic Perception for Forestry*

Conference Publications

under review at ICRA'26 — L. Prior Sancho, T. Belvedere, **M. Jacquet**, M. Tognon, *From Touch to Motion: Direct Tactile Servoing for Human-Guided Aerial Transpostation*

ICRA'25 –Cited: 19 — M. Harms, **M. Jacquet**, K. Alexis, *Safe Quadrotor Navigation using Composite Control Barrier Functions*

ICUAS'25 –Cited: 2 — N. Misyats, M. Harms, M. Nissov, **M. Jacquet**, K. Alexis, *Embedded Safe Reactive Navigation for Multirotors Systems using Control Barrier Functions*

IROS'24 –Cited: 38 — M. Harms, M. Kulkarni, N. Khedekar, **M. Jacquet**, K. Alexis, *Neural Control Barrier Functions for Safe Navigation*

ICRA'24 –Cited: 20 — **M. Jacquet**, K. Alexis, *N-MPC for Deep Neural Network-Based Collision Avoidance exploiting Depth Images*

IROS'22 –Cited: 26 — G. Corsini, **M. Jacquet**, H. Das, A. Afifi, D. Sidobre, A. Franchi, *Nonlinear Model Predictive Control for Human-Robot Handover with Application to the Aerial Case*

References and additional information available upon request

IROS'22 –Cited: 6 — **M. Jacquet**, A. Franchi, *Enforcing Vision-Based Localization using Perception Constrained N-MPC for Multi-Rotor Aerial Vehicles*

AIRPHARO'21 –Cited: 5 — G. Corsini, **M. Jacquet**, AE. Jimenez-Cano, A. Afifi, D. Sidobre, A. Franchi, *A General Control Architecture for Visual Servoing and Physical Interaction Tasks for Fully-actuated Aerial Vehicles*

ICRA'20 –Cited: 28 — **M. Jacquet**, G. Corsini, D. Bicego, A. Franchi, *Perception-constrained and Motor-level Nonlinear MPC for both Underactuated and Tilted-propeller UAVS*

Editorial services

Program Committee — ECMR (2027)

Associate Editor — IROS (2026)

Reviewer — Journals: Nature Communications, T-RO, IJRR, RAL
Conferences: RSS, ICRA, IROS, ICUAS, ECC

Invited Speaker

Nantes, 17/11/2026 — *Learning Representations for Reactive Aerial Navigation*
Talk at ARMEN, LS2N

Rennes, 14/10/2025 — *Learning-based Mapless Navigation for Collision Avoidance*
Talk at Rainbow, Irisa

Rimini, 15/03/2024 — *Learning for Sensor-Based Navigation in Forest Environments*
Talk at ERF24 Workshop *Challenging new applications for autonomous robotics*

Rimini, 15/03/2024 — *Modular Deep Learning for Resilient Navigation Autonomy of Aerial Robots*
Talks at ERF24 Workshops *Navigating the Unknown* and *Aerial robots: Towards Highly Autonomous Drones*

Twente, 25/09/2023 — *Learning Map-free Navigation Policies for Collision Avoidance*
Talk at RAM Group, UTwente

Munich, 02/11/2022 — *Realtime Motor-level NMPC for Multi-rotor ARs Under Perception Constraints*
Talk at Smart Robotics Lab, TUM

06/07/2021 — *Motor and Perception Constrained NMPC for Torque-controlled Generic Aerial Vehicles*
Talk at the Groupe de Travail UAV of the GdR Robotique (remote presentation due to Covid restrictions)

Twente, 22/06/2021 — *Extending the Actuation-Constrained NMPC with Perception Objectives*
Invited Lecturer for the Control for UAVs class at UTwente

Co-supervision of Students

PhD Students

Aug 2026 – Present — Lluis Prior Sancho, INRIA, France
Toward Tactile Sensing for Aerial Physical Interaction (provisory title)
Role: Theoretical supervision

Mar 2023 – Present — Marvin Harms, NTNU, Norway
Safe Quadrotor Navigation in Unknown, Cluttered Environment (provisory title)
Role: Theoretical & technical supervision

Master's Students

Oct 2026 – Present — Andrea Venezia, Polimi, Italy
Enabling MoCap-Free Fly-Crane Operation through Onboard Localization, Relative Perception, and Safety Monitoring (provisory title)
Role: Theoretical & technical supervision

Sep 2024 – Dec 2024 — Nazar Misyats, ENS Rennes, France
Development of a Safe Embedded Drone Controller based on Control Barrier Functions
Role: Topic definition, theoretical supervision, experimental hardware validation
This internship led to a publication to ICUAS

Jan 2024 – Jul 2024	Ask Espensønn Øren, NTNU, Norway *Multi-modal Learning-based Navigation* Role: Topic definition, theoretical & technical supervision
Jan 2024 – Jul 2024	Mathias Skogli, NTNU, Norway *Reinforcement Learning for Autonomous Safe Navigation in Dynamic and Cluttered Environments* *Exploiting Dynamics-Aware Embeddings of Depth Images* Role: Topic definition, theoretical & technical supervision
Jan 2020 – Jul 2021	Max Kivits, University of Twente, The Netherlands *Heterogeneous Cooperative Target Tracking using Nonlinear Model Predictive Control* Role: Topic definition, technical supervision, experimental hardware validation This internship led to a publication to RAL
Jan 2020 – Jul 2020	Vamshi Kodipaka, University of Bourgogne, France *Vision based Multi-robot Tracking of Target with Aerial and Ground Robots* Role: Topic definition, technical supervision

Maintained Open Source Software

Autonomy Stack	Comprehensive Unified Autonomy Stack for Robots – Dev., auth. of related paper – 241 stars
SDF-NMPC	Neural MPC for collision avoidance – Main dev., first auth. of related paper – 55 stars
CCBF	Composite CBF for collision avoidance – Main dev., auth. of related paper – 14 stars
PX4-CBF	Embedded impl. of CCBF in PX4 Autopilot – Dev., auth. of related paper – 13 stars

All my papers also have an open-source implementation for reproducibility (not actively maintained)

Technical skills

Scientific	Neural Networks, Implicit neural representations Nonlinear optimal control Pose estimation, uncertainties, Kalman filtering Deep learning for image and video processing Mathematical modeling, physical parameters estimation
Languages	Python, C++, C, Matlab
Programming	ROS2, ROS, GenoM, PX4, LateX, Git, Docker, Docker-compose
Librairies	PyTorch, OpenCV, Eigen
Hardware	Camera (calibration, data acquisition), Lidars, IMUs, Custom drones & electronics

Languages

French	Native spealer
English	C2, fluent written and oral communication; Toeic obtained in 2015 (940) Professional daily communication at work (working in international teams since 2019)
Spanish	High-school level (B1)
Norwegian	Advanced Beginner
Japanese	Beginner
Italian	Beginner

Various info

Hobbies	Hobby software programming, cycling, brewing, tabletop role-playing, video games, cinema, photography, science-fiction